Testing Evidence Summary

Final report testing section - evidence, results, and open requirements

Scope and provenance

Source: SENG 34213 System Development Project Guideline sections 6.2, 6.4, 10.1, and 10.4; MENTIFY Software Design Specification use cases for login/RBAC, quiz auto-grading, assignment grading, course enrollment, materials, messaging, and analytics.

Test repository branch: feature/TEST-042-final-testing-deliverables; source SHA at PDF generation: fc87f2c. Test case mappings: 663 rows targeting documented JUnit methods.

Test inventory and execution

Route audit inventory: 235 HTTP controller route rows. Source-linked categories: {'REQUEST_WITH_SUCCESS_AND_NEGATIVE_RESPONSE_ASSERTIONS_LINKED': 162, 'REQUEST_WITH_SUCCESS_RESPONSE_ASSERTION_LINKED': 46, 'REQUEST_WITH_NEGATIVE_RESPONSE_ASSERTIONS_ONLY': 27}. Source-linked requests/assertions show test implementation, not that each route passed in a live deployment.

Current local repository verification: 103 JUnit methods; 117 register rows Passed and 546 Not Run, zero unmapped. This is static/local verification, not a live backend integration run. The latest historical live run had 17 methods passed, zero failed/skipped; 23 mapped case rows passed and 640 were not run. That older evidence did not capture a test SHA or GitHub run ID and is not attributed to this PDF's source SHA.

Frontend browser verification: 5 Student/Admin Playwright E2E tests passed. The AI chat spec was intentionally excluded per the team's manual verification decision. The tested E2E files match the current frontend development branch tree; no frontend source change was needed for this task.

Current register input status totals: {'Not Run': 546, 'Passed': 117}. AI endpoint tests were not executed, in keeping with the team's manual-verification decision; the team reports the AI path was manually checked.

Coverage results

Backend JaCoCo source: run local-backend-TEST-042, SHA 8cb633c.

No backend JaCoCo report was imported.

The latest historical backend aggregate is 53.28% line and 33.73% branch across 12 unit modules, below the 80% guideline target and not a changed-code measurement. Backend TEST-042 adds a 90% branch gate for the SDS quiz and assignment grading classes. Full assignment-service verify passed locally with 95 tests; both classes reached 100% line and branch coverage. The branch's Actions artifact is pending.

Performance and staging

Performance: NOT RUN. No approved dedicated target, test account secrets, or successful workflow result was available. No latency/throughput/error measurements are claimed.

Staging: NOT RUN. No approved staging URLs, protected environment approval, or successful smoke workflow result was available.

Completion criteria

- Meet 80% line and branch coverage for changed backend production code and retain CI evidence.
- Retain a successful backend CI run and JaCoCo XML artifact proving the SDS-scoped 90% critical branch gate passed.
- Run the backend integration coverage job and confirm every effective API endpoint has a passing integration case.
- Configure protected performance and staging environments, run their workflows, and retain the evidence artifacts.
- Update the test register, coverage report, performance report, and final report with the actual run IDs, SHAs, outputs, and any remaining blocked cases.